# Driven two-level system beyond the RWA: notes on `task.py`

These notes accompany task.py. Part 1 explains the physics that the program simulates and tests; Part 2 walks through the code, pointing out the non-obvious choices and tricks.

Units: $\hbar = 1$ throughout, so energies and angular frequencies are the same thing. Default parameters of the program: $\varepsilon_0 = 0, \varepsilon_1 = 1, \Omega = 0.1$, initial state $|0\rangle$.

## Part 1 — The physics

### 1.1 The Hamiltonian

$$H(t) = \varepsilon_0 |0\rangle\langle 0| + \varepsilon_1 |1\rangle\langle 1| + 2\Omega\cos(\omega t)\,\big(|0\rangle\langle 1| + |1\rangle\langle 0|\big).$$

With the Pauli matrices in the basis $\big(|0\rangle, |1\rangle\big)$ and $\omega_0 \equiv \varepsilon_1 - \varepsilon_0$ (bare transition frequency):

$$H(t) = \frac{\varepsilon_0 + \varepsilon_1}{2}\,1 - \frac{\omega_0}{2}\,\sigma_z + 2\Omega\cos(\omega t)\,\sigma_x = \frac{\varepsilon_0 + \varepsilon_1}{2}\,1 + \frac{1}{2}\,\mathbf{B}(t)\cdot\boldsymbol{\sigma},$$

$$\mathbf{B}(t) = \big(4\Omega\cos\omega t,\ 0,\ -\omega_0\big).$$

The identity term multiplies the state by a global phase $e^{-i(\varepsilon_0+\varepsilon_1)t/2}$ and has no observable effect: only $\omega_0$, $\Omega$ and $\omega$ matter. The program checks this (`test_global_energy_shift_invariance`).

Note the sign: $|0\rangle$ is the **lower** level when $\varepsilon_1 > \varepsilon_0$, and it sits at the **north** pole of the Bloch sphere ($\sigma_z|0\rangle = +|0\rangle$). The static field therefore points to the south ($B_z = -\omega_0$).

### 1.2 Bloch vector and Bloch equation

For $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$ the Bloch vector is $\mathbf{r} = \langle\boldsymbol{\sigma}\rangle$:

$$x = 2\,\mathrm{Re}(c_0^* c_1), \qquad y = 2\,\mathrm{Im}(c_0^* c_1), \qquad z = |c_0|^2 - |c_1|^2.$$

$z = +1$ means "all population in $|0\rangle$" and $z = -1$ means "all in $|1\rangle$". So $P_1 = (1 - z)/2$.

From the Heisenberg equation $\dot\sigma_j = i[H, \sigma_j]$ and $[\sigma_k, \sigma_j] = 2i\epsilon_{kjl}\sigma_l$:

$$\boxed{\dot{\mathbf{r}} = \mathbf{B}(t) \times \mathbf{r}}$$

This is the equation of a classical magnetic moment in a field $\mathbf{B}$. Two consequences shape every figure:

- $\frac{d}{dt}|\mathbf{r}|^2 = 2\,\mathbf{r}\cdot(\mathbf{B}\times\mathbf{r}) = 0$. A pure state under unitary evolution stays **on the unit sphere**. All the geometry is a curve on $S^2$, and every projection stays inside the unit disc.
- At each instant the motion is an infinitesimal rotation about $\mathbf{B}(t)$, at angular speed $|\mathbf{B}(t)|$.

Without drive ($\Omega = 0$), $\mathbf{B} = -\omega_0\hat z$ and the vector precesses about $z$ at $\omega_0$. Because $\mathbf{B}$ points down, the precession is **clockwise** seen from $+z$: $x = \cos\omega_0 t, y = -\sin\omega_0 t$ when starting from $(1, 0, 0)$ (`test_free_precession`).

## 1.3 A linear drive is two circular drives

The drive field is linear along $x$. Write it as the sum of two counter-rotating circular components:

$$4\Omega\cos\omega t\;\hat{x} = \underbrace{2\Omega(\cos\omega t,\,-\sin\omega t,\,0)}_{\text{co-rotating (clockwise)}} + \underbrace{2\Omega(\cos\omega t,\,+\sin\omega t,\,0)}_{\text{counter-rotating}}.$$

- The **co-rotating** component turns in the same sense as the free precession. Seen by the spin, it is detuned only by $\Delta = \omega_0 - \omega$, so it is nearly resonant when $\omega \approx \omega_0$.
- The **counter-rotating** component turns the other way. Its relative frequency is $\omega + \omega_0 \approx 2\omega_0$, which is far off resonance.

The rotating-wave approximation (RWA) keeps the first component and drops the second. Everything "beyond RWA" in the program comes from the second.

## 1.4 Rotating frame and the RWA: a circle

Go to a frame rotating clockwise about $z$ at the drive frequency $\omega$. For the in-plane components (using $w = x + iy$) this means $w' = w\,e^{i\omega t}$, while $z' = z$. Differentiating $\mathbf{r}' = R_z(\omega t)\,\mathbf{r}$ gives again a Bloch equation, with field

$$\mathbf{B}'(t) = R_z(\omega t)\,\mathbf{B}(t) + \omega\hat{z}.$$

The extra $\omega\hat{z}$ is the "fictitious field" of the rotating frame (the Larmor theorem). Explicitly:

$$\mathbf{B}'(t) = \underbrace{(2\Omega,\;0,\;-\Delta)}_{\mathbf{B}_{\text{eff}}\text{ (static)}} \;+\; \underbrace{2\Omega(\cos 2\omega t,\;\sin 2\omega t,\;0)}_{\text{counter-rotating, now at } 2\omega}.$$

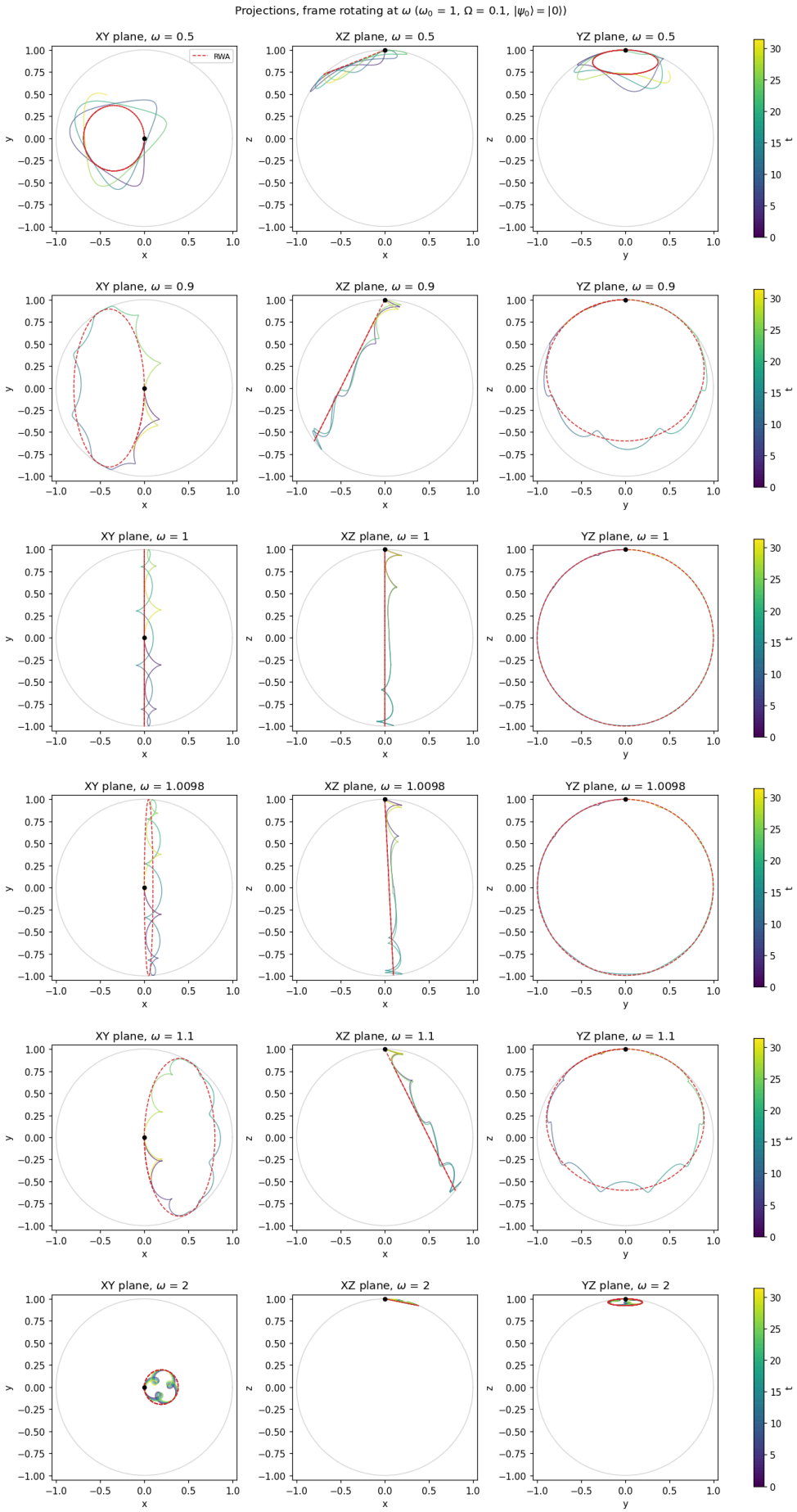
**RWA** means dropping the fast term. Then $\mathbf{B}'$ is static and the solution is a rigid rotation about the fixed axis $\hat{n} = \mathbf{B}_{\text{eff}}/|\mathbf{B}_{\text{eff}}|$, at the generalised Rabi frequency

$$\Omega_R = |\mathbf{B}_{\text{eff}}| = \sqrt{4\Omega^2 + \Delta^2}.$$

**Geometry.** The trajectory is a **circle**: the intersection of the sphere with a cone of axis $\hat{n}$ that passes through the starting point (the north pole). The angle $\theta$ between $\hat{n}$ and $\hat{z}$ satisfies $\cos\theta = -\Delta/\Omega_R$. Rotating $\hat{z}$ by $\pi$ about $\hat{n}$ gives the lowest point

$$z_{\min} = 2n_z^2 - 1 = \frac{\Delta^2 - 4\Omega^2}{\Delta^2 + 4\Omega^2}, \qquad P_1^{\max} = \frac{4\Omega^2}{4\Omega^2 + \Delta^2} \quad \text{(Rabi formula)}.$$

- **Resonance** ($\Delta = 0$): $\hat{n} = \hat{x}$ lies on the equator. The circle is a great circle through both poles, giving complete inversion with period $2\pi/(2\Omega) = \pi/\Omega$.
- **Detuned**: the axis tilts towards a pole. The circle shrinks to a polar cap, so inversion is incomplete but faster ($\Omega_R > 2\Omega$). The sign of $\Delta$ decides whether the cap leans towards $+x$ or $-x$. This is why the $\omega = 0.9$ and $\omega = 1.1$ circles are mirror images in `bloch_projections_rotating.png`.


Projections, frame rotating at ω (ω0 = 1, Ω = 0.1, |ψ0⟩ = |0⟩)
XY plane, ω = 0.5
XZ plane, ω = 0.5
YZ plane, ω = 0.5
RWA
XY plane, ω = 0.9
XZ plane, ω = 0.9
YZ plane, ω = 0.9
XY plane, ω = 1
XZ plane, ω = 1
YZ plane, ω = 1
XY plane, ω = 1.0098
XZ plane, ω = 1.0098
YZ plane, ω = 1.0098
XY plane, ω = 1.1
XZ plane, ω = 1.1
YZ plane, ω = 1.1
XY plane, ω = 2
XZ plane, ω = 2
YZ plane, ω = 2
x
y
z
t

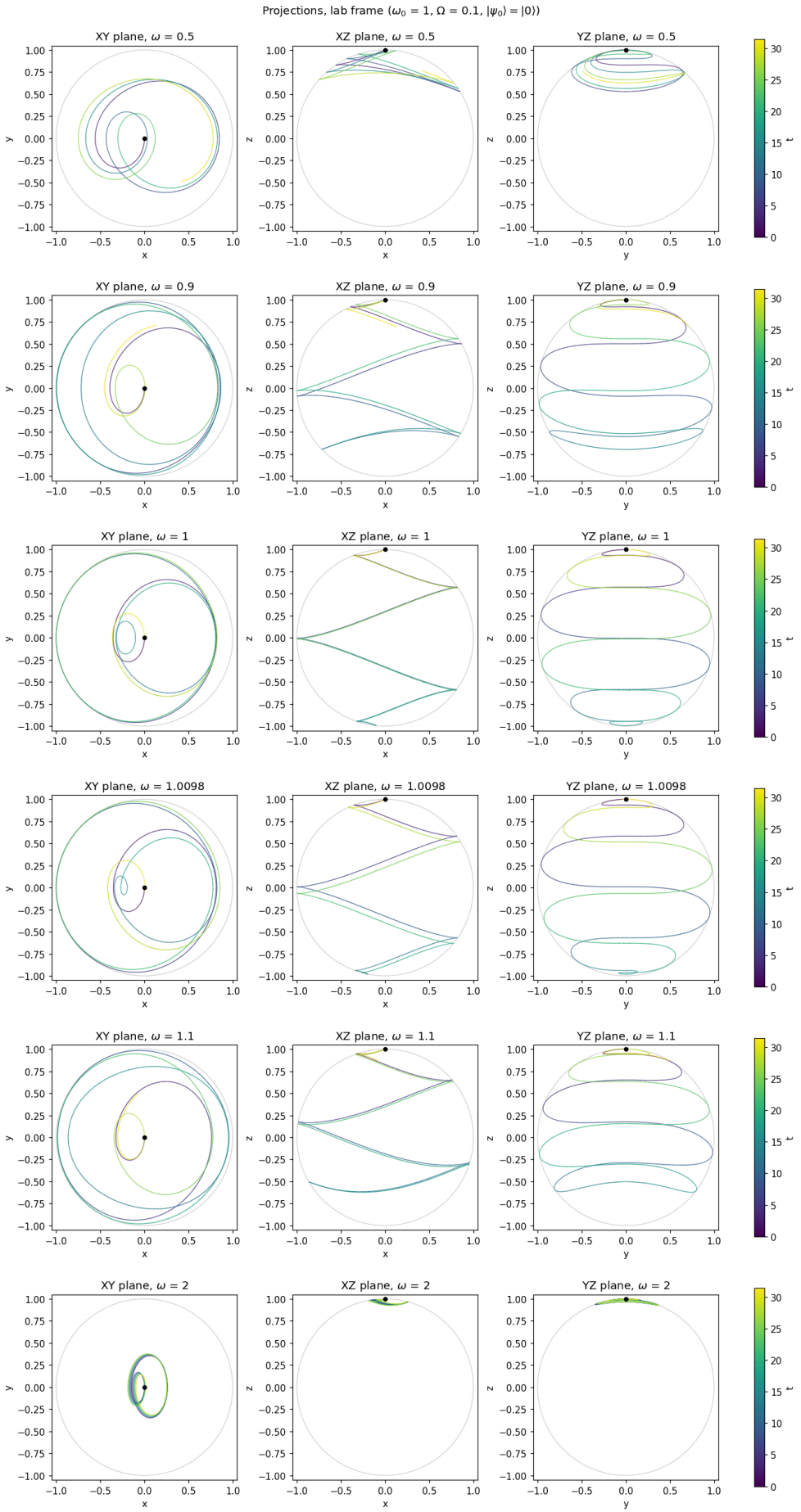
## 1.5 Lab frame: a spiral on the sphere

The lab trajectory is the rotating-frame trajectory turned back clockwise by $\omega t$. On resonance (RWA) the guiding motion is $z = \cos(2\Omega t)$, with the transverse part rotating at $\omega$:

$$x + iy \;\approx\; -i\,\sin(2\Omega t)\,e^{-i\omega t}.$$

- **XY projection**: a spiral that winds out to radius $\sqrt{1-z^2} = 1$ at the equator and winds back in. The two hemispheres project on top of each other, which produces a rosette.
- **XZ / YZ projections**: fast oscillations at $\omega$ inside the envelope $\pm\sqrt{1-z^2}$, which drifts slowly with $z(t)$.
- The number of turns per Rabi cycle is about $\omega/\Omega_R$. Here it is roughly 5, because $\omega/\Omega_R = 1/0.2$.

Physically this is **nutation superposed on precession**. The fast precession at $\omega$ comes from the large static field $\omega_0$ (and the drive locks it to $\omega$). The slow change of polar angle comes from the weak resonant torque $2\Omega$.


Projections, lab frame ($\omega_0$ = 1, Ω = 0.1, $|\psi_0\rangle = |0\rangle$)
XY plane, $\omega$ = 0.5
XZ plane, $\omega$ = 0.5
YZ plane, $\omega$ = 0.5
XY plane, $\omega$ = 0.9
XZ plane, $\omega$ = 0.9
YZ plane, $\omega$ = 0.9
XY plane, $\omega$ = 1
XZ plane, $\omega$ = 1
YZ plane, $\omega$ = 1
XY plane, $\omega$ = 1.0098
XZ plane, $\omega$ = 1.0098
YZ plane, $\omega$ = 1.0098
XY plane, $\omega$ = 1.1
XZ plane, $\omega$ = 1.1
YZ plane, $\omega$ = 1.1
XY plane, $\omega$ = 2
XZ plane, $\omega$ = 2
YZ plane, $\omega$ = 2
x
y
z
t

### 1.6 Beyond RWA (1): micromotion, cusps and the staircase

In the rotating frame the counter-rotating field $2\Omega(\cos 2\omega t, \sin 2\omega t, 0)$ is still there. Linearise around the pole, with $w = x + iy$ and $z \approx 1$:

$$\dot{w} = -i\Delta\, w - 2i\Omega\, e^{2i\omega t} \quad \Rightarrow \quad w_{\text{micro}}(t) = -\frac{2\Omega}{2\omega + \Delta}\, e^{2i\omega t} = -\rho\, e^{2i\omega t}, \qquad \rho = \frac{2\Omega}{\omega + \omega_0}.$$

So on top of the slow RWA circle the vector runs around small circles of radius $\rho$ at frequency $2\omega$. This is **micromotion**: forced response to a far-detuned field, with amplitude equal to field strength divided by detuning.

- The exact curve and the RWA curve both start at the pole, so the exact one is $\rho(1 - e^{2i\omega t})$ away from the RWA one. The maximum distance is therefore $2\rho$ (one diameter). `test_micromotion_amplitude` checks this for $\Omega = 0.01$, $t < 2\pi$ and $\omega = 0.8, 1.0, 1.5$: measured 0.02221, 0.01999, 0.01600 against $2\rho = 0.02222, 0.02, 0.016$. Varying $\omega$ tests the $\omega + \omega_0$ denominator, not just the scaling with $\Omega$.

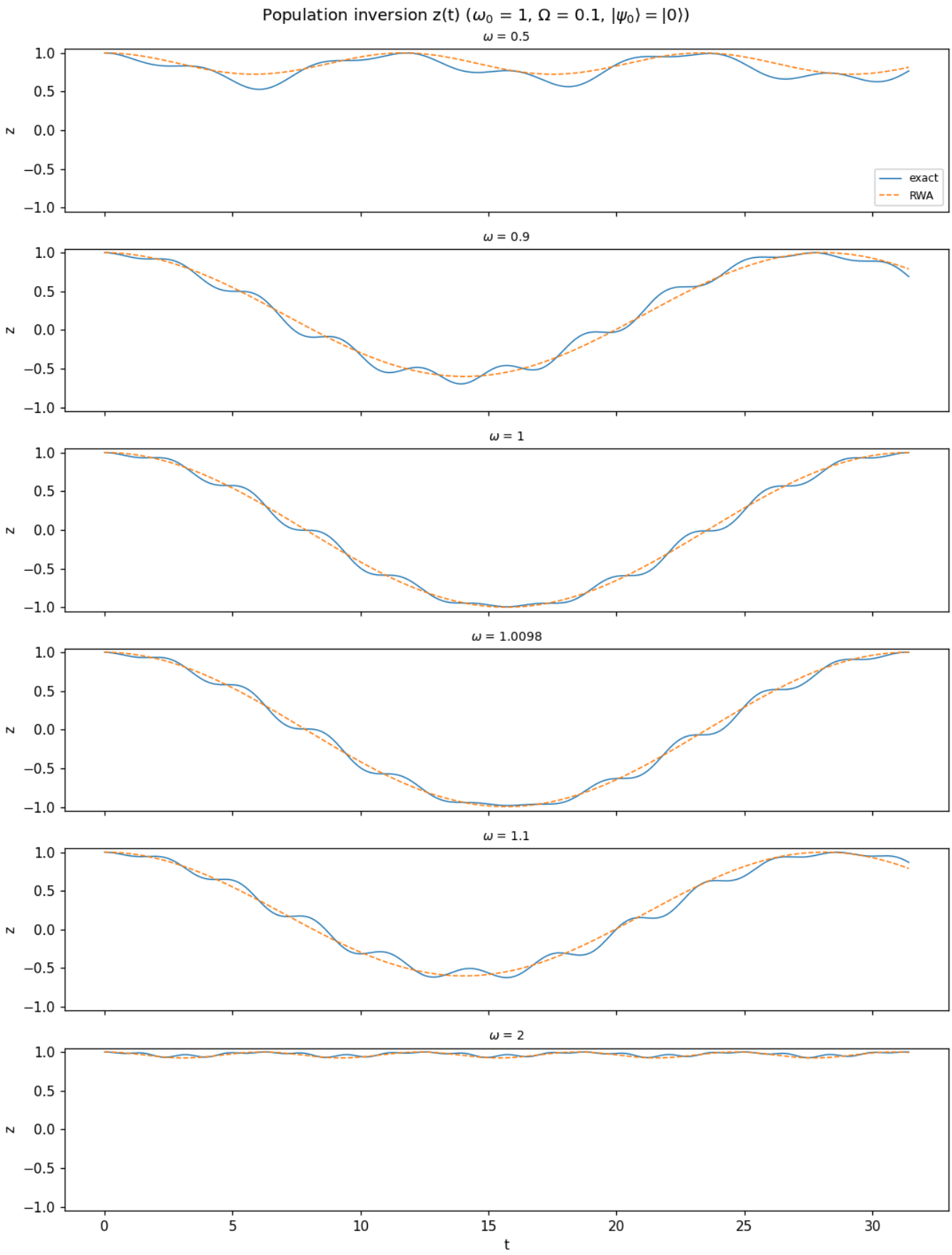
- **Why cusps?** The micromotion speed is $2\omega\rho \approx 2\Omega$ (near resonance). The guiding-centre speed along the RWA great circle is also $\Omega_R \approx 2\Omega$. Two circular motions with comparable speeds add up to a **trochoid**: where the two velocities momentarily cancel, the curve has a cusp or a small loop. That is the scalloped look of the exact curves in the rotating-frame figures.

- The micromotion does not depend on the detuning $\Delta$ (it is set by $\omega + \omega_0$). The RWA circle, by contrast, shrinks like $2\Omega/|\Delta|$ off resonance. So far from resonance (for example $\omega = 0.5$ or $2$) the ripples are comparable to the RWA circle itself, and the RWA is visibly bad (see the table in §1.10).

- **Staircase in $z(t)$** (lab frame). From $\dot{\mathbf{r}} = \mathbf{B} \times \mathbf{r}$:

  $$\dot{z} = B_x\, y - B_y\, x = 4\Omega \cos(\omega t)\, y.$$

  On resonance $y$ itself carries a factor $\cos \omega t$, so $\dot{z} \propto \cos^2 \omega t = \tfrac{1}{2}(1 + \cos 2\omega t)$. The average part is the RWA. The $\cos 2\omega t$ part makes the inversion advance in steps and stall twice per period, whenever the linear field passes through zero. This is clearly visible in `bloch_z_vs_time.png`.


Population inversion z(t) ($\omega_0 = 1$, $\Omega = 0.1$, $|\psi_0\rangle = |0\rangle$)
$\omega = 0.5$
$\omega = 0.9$
$\omega = 1$
$\omega = 1.0098$
$\omega = 1.1$
$\omega = 2$
exact
RWA
z
t
1.0
0.5
0.0
−0.5
−1.0
0
5
10
15
20
25
30

## 1.7 Beyond RWA (2): the Bloch–Siegert shift

The counter-rotating term also has a **secular** (non-oscillating) effect at second order: it shifts the levels. The cleanest way to see this is the dressed-state picture, with the field quantised as photons $n$. Write $2\Omega\cos\omega t = \Omega e^{-i\omega t} + \Omega e^{i\omega t}$. The coupling connects $|0, n\rangle$ with $|1, n \mp 1\rangle$, matrix element $\Omega$ in both cases.

- Resonant pair: $|0, n\rangle \leftrightarrow |1, n-1\rangle$, energy difference $\omega_0 - \omega = \Delta$ (the RWA physics).
- Counter-rotating partners: $|0, n\rangle \leftrightarrow |1, n+1\rangle$ and $|1, n-1\rangle \leftrightarrow |0, n-2\rangle$, both detuned by $\omega_0 + \omega$.

Second-order perturbation theory pushes $|0, n\rangle$ down by $\Omega^2/(\omega_0 + \omega)$ and $|1, n-1\rangle$ up by the same amount. The effective transition frequency becomes

$$\tilde{\omega}_0 = \omega_0 + \frac{2\Omega^2}{\omega_0 + \omega} \;\Rightarrow\; \omega_{\mathrm{res}} \simeq \omega_0 + \frac{\Omega^2}{\omega_0} \qquad \left(= \omega_0 + \Omega_R^2/4\omega_0,\ \Omega_R = 2\Omega\right).$$

This is the **Bloch–Siegert shift** (Bloch & Siegert, 1940). With the default parameters the prediction is $1.0100$. The exact (Floquet) value computed by the program is $\omega_{\mathrm{res}} = 1.00983$; the remaining $1.7 \times 10^{-4}$ is the next order, $O(\Omega^4/\omega_0^3)$.

Geometrically: when you drive at $\omega_0$ "naively", there is a residual detuning $\approx -\Omega^2/\omega_0$. The effective axis is then slightly out of the equatorial plane, and the exact trajectory drifts away from the RWA circle cycle after cycle. Within the default window of one Rabi period, this drift is still small. The `max|r_rot - r_RWA|` value of ≈ 0.19 at $\omega = \omega_0$ is almost entirely the micromotion diameter $2\rho = 0.2$ of §1.6.

## 1.8 Floquet theory: quasi-energies, gap and the stroboscopic circle

The Bloch–Siegert result above is perturbative. The program instead computes the resonance **exactly**, using Floquet theory, which is the natural framework for $H(t+T) = H(t)$ with $T = 2\pi/\omega$.

**Floquet theorem.** There is a basis of solutions of the form

$$|\psi_k(t)\rangle = e^{-i\epsilon_k t}\,|\phi_k(t)\rangle, \qquad |\phi_k(t+T)\rangle = |\phi_k(t)\rangle.$$

The $\epsilon_k$ are the **quasi-energies**. Equivalently, they come from the eigenvalues of the one-period propagator (the monodromy matrix) $U(T) \equiv U(T, 0)$:

$$U(T)\,|\phi_k(0)\rangle = e^{-i\epsilon_k T}\,|\phi_k(0)\rangle.$$

Quasi-energies are only defined **modulo** $\omega$, because $e^{-i(\epsilon+\omega)T} = e^{-i\epsilon T}$ (adding a photon). This is the analogue of crystal momentum in a "Brillouin zone" of width $\omega$.

**Gap.** For a two-level system there are two quasi-energies. Their difference, folded into $[0, \omega/2]$, is the **Floquet gap** $\Delta\epsilon$.

**Why the gap is the exact Rabi frequency.** Since $U(nT) = U(T)^n$:

$$|\psi(nT)\rangle = \sum_k e^{-i\epsilon_k n T}\,|\phi_k\rangle\langle\phi_k|\psi_0\rangle.$$

Any observable sampled stroboscopically, for example $z(nT)$, therefore contains only the frequencies $0$ and $\pm\Delta\epsilon$:

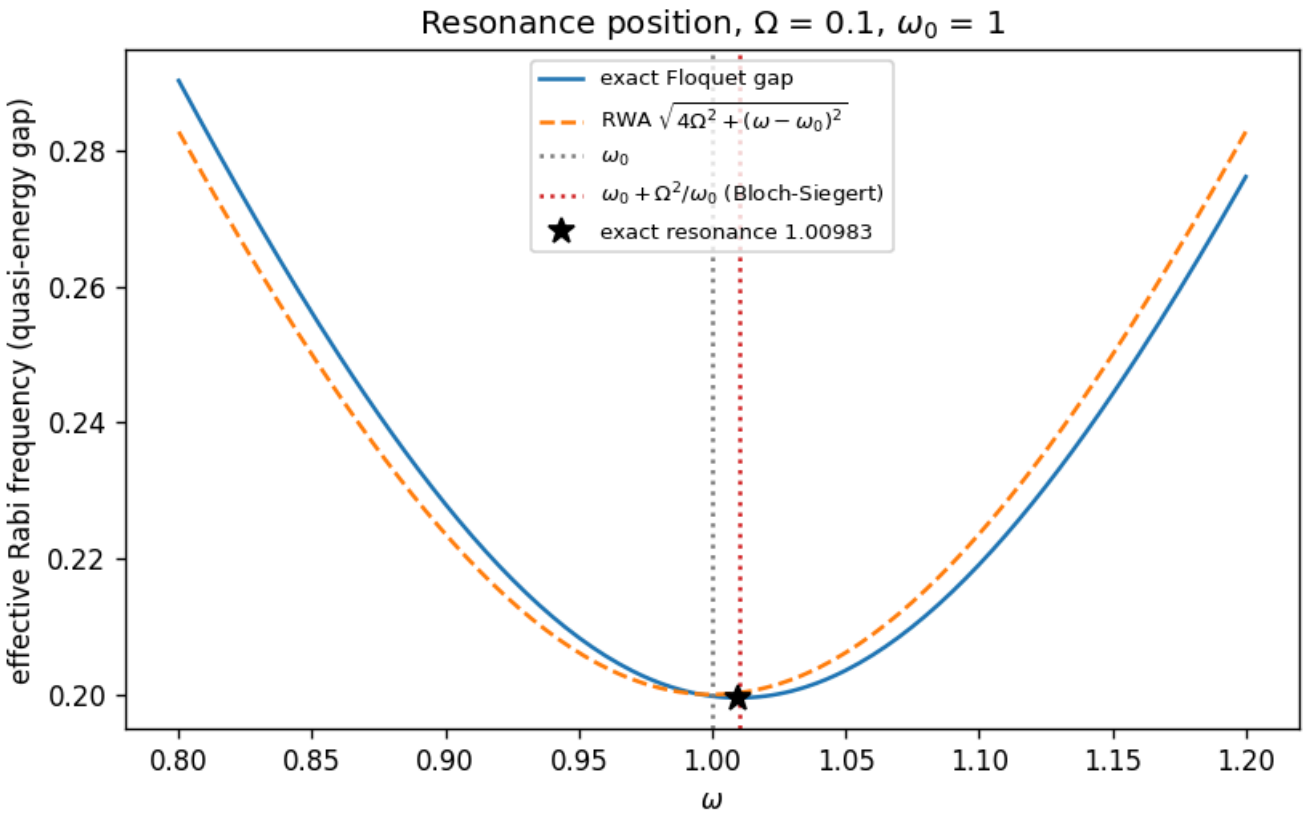
$$z(nT) = a + b\cos(\Delta\epsilon\, nT) + c\sin(\Delta\epsilon\, nT) \qquad \text{exactly.}$$

The modulo-$\omega$ ambiguity does not matter here, since $e^{i\omega nT} = 1$. Equivalently, $U(T) = e^{-iH_F T}$ defines a **time-independent Floquet Hamiltonian** $H_F$. Sampled at $t = nT$, the Bloch vector rotates rigidly about a fixed axis $\hat{n}_F$ at rate $\Delta\epsilon$. So **the stroboscopic points lie exactly on a circle**, which is the exact, non-perturbative version of the RWA circle. (At $t = nT$ the lab and rotating frames coincide.) `test_bloch_siegert_shift` verifies this to $10^{-6}$.

**Resonance = minimum gap.** Near the one-photon resonance the two Floquet levels have an avoided crossing:

$$\Delta\epsilon(\omega) \approx \sqrt{\Omega_{R,\min}^2 + (\omega - \omega_{\text{res}})^2}.$$

In the RWA, $\Delta\epsilon = \sqrt{4\Omega^2 + (\omega - \omega_0)^2} = \Omega_R$, which is minimal at $\omega_0$. The exact gap has its minimum shifted to $\omega_{\text{res}}$, the Bloch–Siegert-shifted resonance. `bloch_siegert.png` shows both curves. The minimum value ( $0.19950$ vs $2\Omega = 0.2$) is the exact effective Rabi frequency on resonance.




**Why the gap and not "maximum inversion".** The quasi-energies do not depend on where the period starts. The propagator over $[t_0, t_0 + T]$ is $U(t_0, 0)\, U(T)\, U(t_0, 0)^\dagger$, which has the same eigenvalues. The axis $\hat{n}_F$, however, **does** depend on the drive phase at $t = 0$: it tilts by $O(\Omega/\omega)$. On top of that, the instantaneous $z(t)$ includes the micromotion. As a result, the deepest $z$ reached from $|0\rangle$ in a finite window changes by fractions of a percent with $\omega$ and window length, and is not a clean resonance marker at the Bloch–Siegert level. The gap is gauge-invariant, so the program uses the gap.

In this program the drive phase is fixed by $\cos(\omega t)$, i.e. the field is at a maximum at $t = 0$.

### 1.9 Far from resonance and multi-photon resonances

- **Slow drive** ($\omega \ll \omega_0$): the field direction swings slowly in the XZ plane. The vector makes small fast precession circles (at $\approx |\mathbf{B}|$) about the instantaneous field and follows it adiabatically. The trajectory stays near the pole and is elongated along $x$. The case $\omega = 0.5$ is intermediate: the RWA circle and the micromotion have similar sizes.

- **Fast drive** ($\omega \gg \omega_0$): the field averages out over a period. $\mathbf{r}$ only wobbles around the pole ($\omega = 2$ in the figures: a small cap decorated with $2\omega$ loops).

- **Only odd multi-photon resonances exist.** $H$ has a *generalised parity* symmetry, a half-period shift combined with $\sigma_z$:

  $$\sigma_z \, H(t + T/2) \, \sigma_z = H(t),$$

  because $\cos(\omega t + \pi) = -\cos \omega t$ and $\sigma_z \sigma_x \sigma_z = -\sigma_x$. In the dressed basis $|0, n\rangle$ has parity $(-1)^n$ and $|1, m\rangle$ has parity $-(-1)^m$. A $k$-photon resonance connects $|0, n\rangle$ and $|1, n-k\rangle$, which have the same parity only if $k$ is **odd**. For even $k$ the levels cross exactly instead of anticrossing, and there is no resonant population transfer.

- **Where the multi-photon resonances sit.** The same second-order argument as in §1.7 applies, but now *both* one-photon couplings are off resonance. $|0, n\rangle$ is pushed down by $\Omega^2/(\omega_0 - \omega) + \Omega^2/(\omega_0 + \omega)$ and $|1\rangle$ is pushed up by the same amount. The $k$-photon condition $k\omega = \tilde{\omega}_0$, evaluated at $\omega \approx \omega_0/k$, gives

  $$\omega_3 \simeq \frac{\omega_0}{3} + \frac{3\Omega^2}{2\omega_0}, \qquad \omega_2 \simeq \frac{\omega_0}{2} + \frac{8\Omega^2}{3\omega_0}.$$

  (For $k = 1$ the $\omega_0 - \omega$ term is the resonant coupling itself and drops out, which gives back the Bloch–Siegert formula.)

Two tests check all of this with $\Omega = 0.1$:

| | **3-photon** (`test_three_photon_resonance`) | **2-photon** (`test_no_two_photon_resonance`) |
|---|---|---|
| second-order estimate | $\omega_3 = 0.3483$ | $\omega_2 = 0.5267$ |
| exact minimum of the Floquet gap | $0.3480$ | $0.5261$ |
| gap at the minimum | $4.2 \times 10^{-3}$ (**avoided** crossing) | $6 \times 10^{-9}$ (**exact** crossing, integrator noise) |
| best stroboscopic transfer $P_1^{\max}$ | $1.0000$ (at $\omega = 0.3486$) | $\leq 0.262$ in $\omega_2 \pm 0.02$, no peak |

On the 2-photon side the levels *are* resonant (the gap closes), but they are not coupled. The background $P_1^{\max} \approx 0.25$ is not resonant transfer. It comes from dressing: at this fairly strong, off-resonant drive the Floquet states at $t = 0$ are tilted by about 30° from $|0\rangle$, $|1\rangle$. With $\Omega = 0.05$ it drops to about 0.07.

As a sanity check that the parity test can actually fail, I added a small static $0.05\,\sigma_x$ term, which breaks the symmetry. The 2-photon crossing then opens to an avoided crossing (gap $7 \times 10^{-3}$) and `test_no_two_photon_resonance` fails, while the 3-photon test still passes.

### 1.10 What the default run shows ($\omega_0 = 1$, $\Omega = 0.1$)

| $\omega$ | $\Delta$ | $\Omega_R$ (RWA) | $z_{\min}$ RWA | $z_{\min}$ exact | max $\lvert\mathbf{r}_{\rm rot} - \mathbf{r}_{\rm RWA}\rvert$ | max $\lvert\lvert\mathbf{r}\rvert - 1\rvert$ |
|---|---|---|---|---|---|---|
| 0.5 | 0.5 | 0.5385 | 0.7241 | 0.5283 | 0.316 | 7e-11 |
| 0.9 | 0.1 | 0.2236 | -0.6000 | -0.6964 | 0.255 | 3e-10 |
| 1.0 | 0 | 0.2000 | -1.0000 | -0.9949 | 0.194 | 3e-10 |
| 1.0098 | -0.0098 | 0.2002 | -0.9952 | -0.9810 | 0.194 | 3e-10 |
| 1.1 | -0.1 | 0.2236 | -0.6000 | -0.6223 | 0.241 | 3e-10 |
| 2.0 | -1.0 | 1.0198 | 0.9231 | 0.9252 | 0.141 | 3e-11 |

Reading guide:

- The last column is the integrator error. Physically $|\mathbf{r}| = 1$ exactly.
- 0.9 and 1.1 are symmetric in the RWA but not in reality, and the asymmetry is a good example of §1.8. I measured it separately with stroboscopic sampling over 3000 periods; it is not part of the program.
  - **Floquet gap:** 0.2280 at $\omega = 0.9$ and 0.2191 at $\omega = 1.1$ (RWA: 0.2236 for both). So 1.1 is *closer* to the true resonance, as the Bloch–Siegert shift to 1.0098 predicts.
  - **Depth from $|0\rangle$:** the exact stroboscopic circle still reaches *deeper* at 0.9 ($z_{\min} = -0.697$) than at 1.1 ($-0.504$).
  - **Why:** with the drive $\propto \cos\omega t$, the micromotion at $t = 0$ is $w_{\rm micro}(0) = -\rho$. So the micromotion-free "guiding centre" starts not at the pole but displaced by $\rho \approx 0.1$ rad (≈ 6°) towards $+x$. That displacement moves the effective start away from the rotation axis for $\Delta > 0$ (larger circle) and towards it for $\Delta < 0$ (smaller circle).
  - **Check:** simple cone geometry with this 6° displacement gives roughly $-0.67$ and $-0.50$, close to the exact values.
  - **In the table:** the lab-frame $z_{\min}$ column also includes the instantaneous micromotion between stroboscopic points. That is why it differs again from the stroboscopic values.
- The $z_{\min}$ columns are taken over a single window $t \in [0, \pi/\Omega]$. Differences of order $10^{-2}$ near resonance reflect the micromotion and the phase dependence discussed in §1.8, not the resonance position.

# Part 2 — The code

## 2.1 Layout

| Section | Functions | Role |
|---|---|---|
| physics | `hamiltonian`, `evolve_states`, `bloch_vector` | exact dynamics |
| | `bloch_field`, `bloch_equation_trajectory` | independent classical cross-check |
| | `to_rotating_frame`, `rwa_trajectory_rotating`, `rwa_summary` | rotating frame and analytic RWA |
| | `one_period_propagator`, `floquet_quasienergies`, `stroboscopic_max_transfer`, `floquet_gap`, `find_resonance` | Floquet analysis: exact resonances (Bloch–Siegert, multi-photon) |
| | `simulate` | runs everything for a list of $\omega$ |
| plots | `plot_3d`, `plot_projections`, `plot_z_vs_time`, `plot_bloch_siegert` | figures |
| tests | `test_*`, `run_tests` | verification |
| CLI | `main` | arguments, run, save |

Only `numpy`, `scipy` and `matplotlib` are needed. `matplotlib` is imported only inside `main`, so the physics and the tests can be imported (for example by `pytest`) without a display.

## 2.2 Time evolution: why `solve_ivp`, not `expm`

The earlier exercises (the model.py style) used $U = e^{-iHt}$ via `scipy.linalg.expm`. That formula is **wrong** for a time-dependent $H$, because $H(t)$ and $H(t')$ do not commute: the true propagator is a time-ordered exponential. `evolve_states` integrates $i\,\dot{\psi} = H(t)\psi$ directly:

```
sol = solve_ivp(rhs, (t0, t1), psi0_complex, t_eval=t_eval,
                method="DOP853", rtol=1e-10, atol=1e-12)
```

Points worth noting:

- **Complex state vector.** `solve_ivp`'s explicit Runge–Kutta methods (`RK45`, `RK23`, `DOP853`) accept complex `y0` directly. There is no need to split into real and imaginary parts. (The implicit methods `Radau`/`BDF`/`LSODA` would need splitting.)
- **DOP853** is an 8th-order Dormand–Prince method. For smooth, non-stiff oscillatory problems it reaches $10^{-10}$ with far fewer steps than RK45. The norm is not conserved exactly by construction (RK is not unitary), but the drift is $\sim 10^{-10}$ (last column of the table) and is tested.
- **`t_eval` does not set the step size.** The solver picks its own adaptive steps. Values at the requested times come from DOP853's 7th-order dense-output interpolant. So `--dt` only controls the plotting resolution, not the accuracy.
- `sol.y` has shape `(2, N)`. The function returns `sol.y.T`, shape `(N, 2)`, so that `psi[n]` is the state at `t[n]`.

Alternatives would be piecewise-constant `expm` products, or a Magnus expansion, which is exactly unitary. They are not needed at this accuracy.

### 2.3 Bloch vector, vectorised

```
c0, c1 = psi[..., 0], psi[..., 1]
rho01 = np.conj(c0) * c1
return np.stack([2*rho01.real, 2*rho01.imag, abs(c0)**2 - abs(c1)**2], axis=-1)
```

- The ellipsis `...` makes the same function work for a single state (shape `(2,)`) and for a whole trajectory (shape `(N, 2)`), with no Python loop.
- It uses the closed forms of §1.2 instead of computing $\langle\psi|\sigma_j|\psi\rangle$ with three matrix products. `rho01` is the off-diagonal density-matrix element $\rho_{01} = c_0^* c_1$.
- Global phases cancel automatically in $c_0^* c_1$ and $|c_i|^2$. This is why the identity part of $H$ needs no special treatment.

### 2.4 An independent check: the classical Bloch equation

`bloch_field` and `bloch_equation_trajectory` integrate $\dot{\mathbf{r}} = \mathbf{B}(t) \times \mathbf{r}$ as a **real 3-vector** ODE, using `np.cross`. This is a different equation, in a different space, with no wavefunction. `test_schrodinger_vs_bloch_equation` compares it with the Schrödinger result for a generic initial state. That test catches sign or factor errors in the Hamiltonian, in the Bloch-vector formulas, or in the field $\mathbf{B}$. A single-method check cannot catch those, because such errors are self-consistent.

### 2.5 Rotating frame with complex numbers

```
w = (r[:, 0] + 1j*r[:, 1]) * np.exp(1j*omega*t)
```

A rotation about $z$ is multiplication of $x + iy$ by a phase. This line is the whole frame change $R_z(\omega t)$, applied to all times at once. The sign $+i\omega t$ undoes the **clockwise** motion. With the opposite sign, the resonant trajectory would rotate at $2\omega$ instead of standing still. A quick sanity check: on resonance the exact rotating-frame curve follows the static RWA circle.

### 2.6 Analytic RWA: Rodrigues' rotation formula

`rwa_trajectory_rotating` does not integrate anything. A static field means a rigid rotation by angle $\Omega_R t$ about $\hat{n}$ :

$$\mathbf{r}(t) = \mathbf{r}_0 \cos\Omega_R t + (\hat{n} \times \mathbf{r}_0)\sin\Omega_R t + \hat{n}(\hat{n}\cdot\mathbf{r}_0)(1 - \cos\Omega_R t).$$

```
c = np.cos(omega_r*t)[:, None]          # shape (N, 1)
s = np.sin(omega_r*t)[:, None]
return r0*c + np.cross(n, r0)*s + n*np.dot(n, r0)*(1 - c)   # (N, 3)
```

- `[:, None]` turns the time arrays into columns. NumPy broadcasting then combines `(N,1)` with the 3-vectors `(3,)` into an `(N,3)` trajectory.
- The guard `if omega_r == 0` covers $\Omega = 0$, $\Delta = 0$, where $\hat{n}$ is undefined (no rotation at all).
- `rwa_summary` uses `np.hypot(2*Omega, delta)` for $\Omega_R$, which avoids overflow and underflow in `sqrt(a**2 + b**2)`. It also returns $z_{\min}$ from §1.4.

## 2.7 Floquet machinery

**Monodromy matrix** (`one_period_propagator`, `floquet_quasienergies`):

```
U = np.column_stack([evolve_states(e, [0, T], ...)[-1] for e in np.eye(2)])
return -np.angle(np.linalg.eigvals(U)) / T
```

- Evolving each basis vector for one period and stacking the results as columns builds $U(T)$ column by column: column $j$ is $U(T)|j\rangle$.
- Its eigenvalues are $e^{-i\epsilon T}$, so $\epsilon = -\arg(\lambda)/T$. `np.angle` returns values in $(-\pi, \pi]$, so the quasi-energies come out already in the first zone $(-\omega/2, \omega/2]$.
- `eigvals`, not `eig`: only the eigenvalues are needed, and eigenvectors of nearly degenerate Floquet states are numerically ill-conditioned (I hit exactly this problem in the even-photon check of §1.9).

**Best stroboscopic transfer without eigenvectors** (`stroboscopic_max_transfer`). Any $U \in U(2)$ can be written as $U = e^{i\varphi}(\cos a\, 1 - i \sin a\, \hat{n}\cdot\boldsymbol{\sigma})$. Sampled at $t = nT$, the Bloch vector rotates about $\hat{n}$, so starting from the north pole the best reachable transfer is $\sup_n P_1(nT) = 1 - n_z^2$. Reading off $\sin a\, n_z$ and $\sin a \sqrt{n_x^2 + n_y^2}$ from the matrix elements:

$$P_1^{\max} = \frac{|U_{10}|^2}{|U_{10}|^2 + \tfrac{1}{4}|U_{11} - U_{00}|^2}.$$

The global phase $e^{i\varphi}$ cancels in the ratio, so no $\sqrt{\det U}$ normalisation is needed. The formula stays well conditioned near crossings, where eigenvectors do not. The only singular point is $U \propto 1$, where the function returns 0. One evaluation costs one period of integration, compared with thousands of periods for a brute-force search. `test_stroboscopic_max_transfer_formula` compares it with an explicit $\max_n |\langle 1|U^n|0\rangle|^2$ over 20 000 periods.

**Folding the gap** (`floquet_gap`):

```
d = (e[0] - e[1]) % omega
return min(d, omega - d)
```

The difference is only defined modulo $\omega$, and its sign is arbitrary (eigenvalue order is not fixed). `% omega` maps it to $[0, \omega)$. Then `min(d, omega-d)` picks the representative in $[0, \omega/2]$. The result is a continuous function of $\omega$ near the resonance, which the optimiser needs.

**Finding the resonance** (`find_resonance`):

```
minimize_scalar(gap, bounds=(omega0 - 2*Omega, omega0 + 2*Omega),
                method="bounded", options={"xatol": 1e-8})
```

- Brent's bounded method needs no derivative. Each evaluation costs two one-period integrations.
- The window $\pm 2\Omega$ contains the anticrossing (the shift is $\Omega^2/\omega_0 \ll 2\Omega$ for a weak drive) and stays away from the folding kinks at $\Delta\epsilon = \omega/2$.
- **Accuracy caveat.** The minimum is quadratic, so a numerical error $\delta$ in the gap gives an uncertainty of about $\sqrt{2\Omega_R \delta}$ in the position. With $\delta \sim 10^{-10}$ that is $\sim 10^{-5}$: `xatol=1e-8` overstates the real precision. This is irrelevant compared with the shift itself ($10^{-2}$).

## 2.8 Driver: `simulate`

- `np.arange(0, t_max + dt/2, dt)` includes `t_max` without the floating-point risk of `arange(0, t_max + dt, dt)`, which sometimes produces one extra point.
- For each $\omega$ it stores the lab trajectory, its rotating-frame version and the analytic RWA curve in a dict. The plotting functions pick `"lab"` or `"rot"` via the `frame` argument, so the same plotting code serves both frames.
- The default $\omega$ list includes $\omega_{\mathrm{res}}$ from `find_resonance`, so the exact resonance is always simulated.

## 2.9 Plotting tricks

- **Time-coloured curves** (`plot_projections`). Matplotlib cannot colour a single line by a parameter, so the curve is cut into segments:

  ```
  pts = np.column_stack([r[:, a], r[:, b]]).reshape(-1, 1, 2)          # (N,1,2)
  segs = np.concatenate([pts[:-1], pts[1:]], axis=1)                   # (N-1,2,2)
  lc = LineCollection(segs, cmap="viridis"); lc.set_array(t[:-1])
  ```

  Each segment gets the colour of its starting time. This is what makes the overlapping lab-frame spirals readable.

- `constrained_layout=True` and one `colorbar` per row (`ax=axes[i, :].tolist()`) keep the colour bars from overlapping the panels. `tight_layout` cannot handle multi-axes colour bars.

- `ax.set_aspect("equal")` with limits $\pm 1.05$ keeps the unit circle round, so you can see that projections stay inside it.

- **Sphere wireframe** (`_draw_sphere`): `np.mgrid[0:2*np.pi:30j, 0:np.pi:15j]`. An imaginary step `30j` in `mgrid` means "30 points, endpoints included", like `linspace`, not "step 30".

- `set_box_aspect((1,1,1))` makes the 3-D sphere look round. It sits inside `try/except AttributeError` because it only exists in Matplotlib ≥ 3.3.

- `plt` is passed into the plotting functions instead of being imported at module level. Together with `matplotlib.use("Agg")` in `main`, called **before** importing `pyplot`, this lets the script save figures without a display, while `--show` keeps the interactive backend.

## 2.10 The tests and what each one proves

| Test | Checks | Why it is meaningful |
|---|---|---|
| `test_bloch_vector_stays_on_unit_sphere` | $\lvert\lvert\mathbf{r}\rvert - 1\rvert < 10^{-8}$ for 3 drive frequencies | unitarity / purity; also bounds the integrator error |
| `test_free_precession` | $\Omega = 0$, start on equator, compare to $(\cos\omega_0 t, -\sin\omega_0 t, 0)$ | fixes the **sign** conventions (clockwise precession, $B_z = -\omega_0$) against an exact solution |
| `test_schrodinger_vs_bloch_equation` | quantum vs classical integration, generic $\psi_0$ with complex phase | independent derivation (§2.4); a non-trivial $\psi_0$ exercises $x$, $y$ and $z$ |
| `test_global_energy_shift_invariance` | $(\varepsilon_0, \varepsilon_1) = (0, 1)$ vs $(5, 6)$ give the same $\mathbf{r}(t)$ | only $\omega_0$ is physical (§1.1) |
| `test_rwa_limit` | deviation from RWA over one Rabi period, for $\Omega = 0.02$ and $0.01$: small, and the ratio is in $[0.35, 0.65]$ | checks the **scaling** $\propto \Omega/\omega_0$ (micromotion $2\rho$ and Bloch–Siegert dephasing $\sim \pi\Omega$ are both linear in $\Omega$), not just "small" |
| `test_bloch_siegert_shift` | $\omega_{\text{res}}$ within 10% of the shift $\Omega^2/\omega_0$; gap within 2% of $2\Omega$; stroboscopic $z(nT)$ fits $a + b\cos + c\sin$ at the Floquet gap to $10^{-6}$, with amplitude $> 0.9$ | connects the Floquet calculation to the actual dynamics: the gap **is** the Rabi frequency (§1.8) |
| `test_micromotion_amplitude` | on a short window, max $\lvert\mathbf{r}_{\text{rot}} - \mathbf{r}_{\text{RWA}}\rvert = 4\Omega/(\omega + \omega_0)$ within 2%, for $\omega = 0.8, 1.0, 1.5$ | the size of the counter-rotating effect, including its $\omega + \omega_0$ dependence (§1.6) |
| `test_stroboscopic_max_transfer_formula` | closed-form $P_1^{\max}$ vs explicit $\max_n \lvert\langle 1\rvert U^n\lvert 0\rangle\rvert^2$ over 20 000 periods | validates the helper the next two tests rely on (§2.7) |
| `test_three_photon_resonance` | gap minimum within 10% of the shift $3\Omega^2/2\omega_0$; gap finite ($10^{-3}$–$10^{-2}$); $P_1^{\max} > 0.999$ nearby | odd resonances are avoided crossings with full transfer (§1.9) |
| `test_no_two_photon_resonance` | gap closes ($< 10^{-6}$) within 10% of the shift $8\Omega^2/3\omega_0$; $P_1^{\max} < 0.5$ over $\pm 0.02$ around it | the parity selection rule: resonant but uncoupled. A mutation check (symmetry-breaking $\sigma_x$ term) makes this test fail (§1.9) |

Design choices behind the tests:

- **Convergence-order test instead of a hard threshold** (`test_rwa_limit`). Halving $\Omega$ should halve the error. A ratio test is insensitive to the unknown prefactor, and it fails if the error is accidentally $O(1)$ or $O(\Omega^2)$.
- **Stroboscopic least-squares fit** (`test_bloch_siegert_shift`). By §1.8, $z(nT)$ is *exactly* a constant plus one sinusoid of frequency $\Delta\epsilon$. A linear fit with basis $\{1, \cos\Delta\epsilon\, t_n, \sin\Delta\epsilon\, t_n\}$ (`np.linalg.lstsq`) leaves a residual at the integrator-noise level only if $\Delta\epsilon$ is right. The amplitude condition $> 0.9$ makes sure there is an actual near-complete oscillation to fit.

- The tests use the plain `assert` style and the `test_` prefix. They run in two ways: from `python task.py`, through `run_tests`, before plotting; and from `python -m pytest task.py`. Naming the file explicitly makes pytest collect it even though it is not called `test_*.py`.

## 2.11 Command line

```
python task.py [--eps0 E0] [--eps1 E1] [--Omega W] [--omegas w1 w2 ...]
               [--t-max T] [--dt DT] [--show] [--no-tests]
```

- `argparse` takes its description from the second line of the module docstring (`__doc__.splitlines()[1]`), so the help text and the docstring stay in sync.
- `args.omegas or [...]` and `args.t_max or np.pi/Omega` fill in physics-based defaults. By default the time window is one resonant Rabi period $\pi/\Omega$.
- Figures are written next to the script (`HERE`), whatever the current directory is.
- Limitations: the initial state is fixed to $|0\rangle$ from the command line (`simulate` accepts `psi0` if you call it from Python), and the drive phase is fixed to $\cos \omega t$ (§1.8).